

- (c) The cookies are then packed into a box in the shape of a regular hexagon which can contain 7 cookies.
The cross-section of the box is as shown in Figure 1.

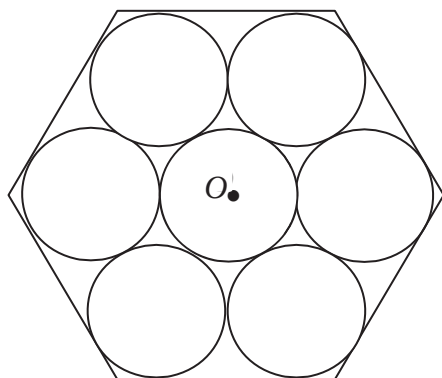


Figure 1

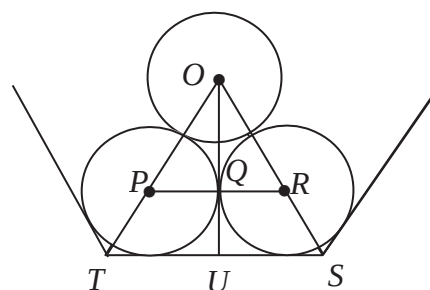


Figure 2

Three of the cookies are shown in Figure 2.

O is the centre of the hexagonal box and P and R are the centres of two cookie as shown in Figure 2. Q is the point where the two cookies touch and U is the midpoint of TS , a side of the box.

- (i) Calculate the length of OP .

Answer _____ cm [1]

- (ii) Calculate the length of OQ .

Answer _____ cm [2]

- (iii) Calculate the length of one side of the box.

Answer _____ cm [2]

- 11 The interior of a chest freezer as shown in Figure A can be modelled as a cuboid with length and height of 100 cm and a width of 55 cm as shown in Figure B.



Figure A

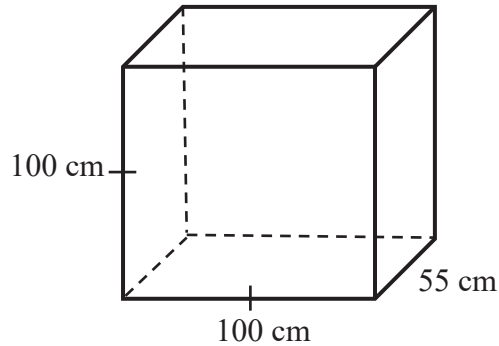


Figure B

Cool air must be circulated inside the freezer at any time.

When cool air meets the warm and humid air that is outside of the freezer, frost is formed at the point of contact.

To allow the freezer to work efficiently, defrosting is required when the layer of frost exceeds 1 cm.

- (a) Due to the opening and closing of the freezer door, frost of x cm is formed uniformly on the four vertical walls of the freezer. The area of the horizontal base of the freezer not covered by the frost is $105(50 - x)$ cm².

Explain if defrosting is required and show your calculations clearly.

Answer _____

_____ [4]

- (b) Kara wants to design a box that can contain the freezer for shipping. There must be free space between the freezer and the box to allow for ease of removal and addition of protective materials to prevent scratches.

It is recommended to have a minimum distance of 2 cm of free space around the sides and a minimum distance of 3 cm of free space at the top. The thickness of the freezer wall and door of the freezer is 3.5 cm.

Find the smallest dimensions of the box required. State one assumption made in your calculation.

Answer Length = _____ cm [1]

Height = _____ cm [1]

Width = _____ cm [1]

Assumption : _____

_____ [1]

END OF PAPER